Roll No.

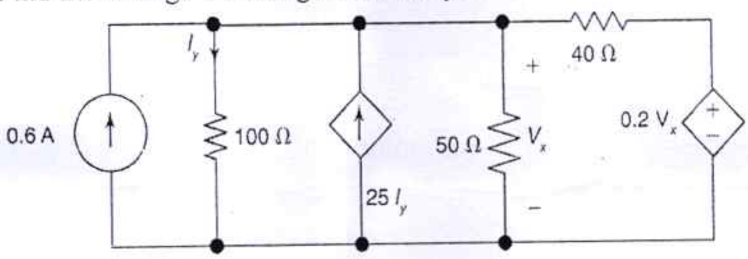
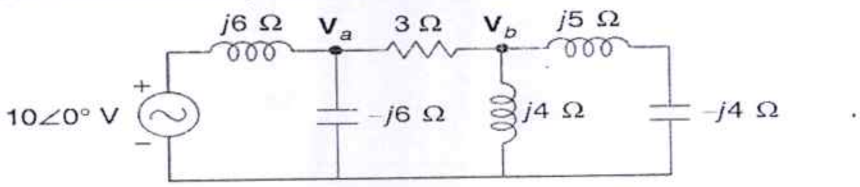
ODD END SEMESTER EXAMINATION DECEMBER - 2025

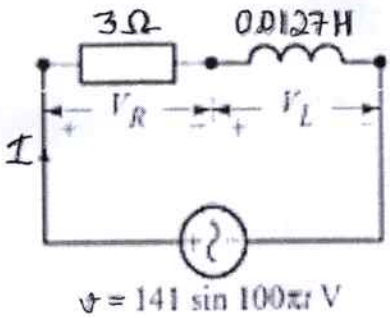
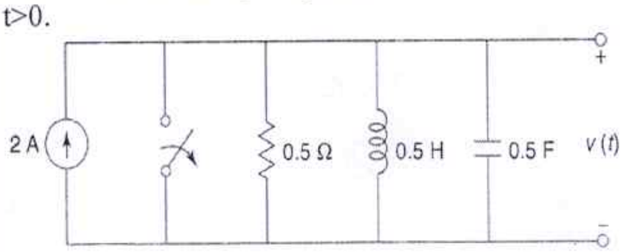
Name of the Program: B. Tech. (EE) Semester: 3rd

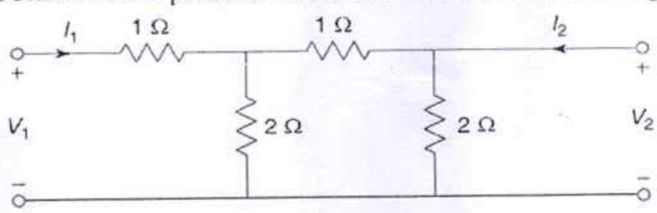
Name of the Course: Electrical Circuit Analysis. Course Code: - TEE 301

Time: 3 hrs Maximum Marks: 100

Note
(i) All questions are compulsory.
(ii) Answer any two sub questions among a, b &c in each main question
(iii) Total marks in each main question are twenty.
(iv) Each question carries 10 marks

Q. 1	(2×10 Marks)	
(a)	Find the voltage Vx using nodal analysis 	CO 1, CO 2, CO3
	OR	
(b)	In the network shown, determine Va and V_b 	CO 1, CO 2, CO3
	OR	
(c)	Explain with illustrative examples the meaning of the following terms (i) Incidence matrix (ii) Tie-set Matrix (iii) Cut-set matrix	CO 1, CO 2, CO 3
Q. 2	(2×10 Marks)	

(a)	For the series RL circuit shown in Fig. (i) Calculate the rms value of the steady state current and the relative phase angle (ii) Write the expression for the instantaneous current (iii) Find the average power dissipated in the circuit (iv) Determine the power factor (v) Draw the phasor diagram  	CO 1, CO 2, CO3, CO4
	OR	
(b)	Derive the relationship between line voltage and phase voltage in a star-connected system. Illustrate your derivations with phasor diagrams.	CO 1, CO 2, CO3, CO4
	OR	
(C)	Solve (i) $y'' + 4y = \delta(t),\ y(0) = 0,\ y'(0) = 0.$ (ii) $y'' + 3y' + 2y = t\delta(t-1),\ y(0) = 0,\ y'(0) = 0.$	CO 2, CO3, CO5
Q. 3	(2×10 Marks)	
(a)	The switch in the fig is opened at time t=0. Determine the voltage v(t) for t>0.  	CO 2, CO3, CO5
	OR	
(b)	Obtain the pole-zero plot of the following functions (a) $F(s) = \frac{s(s+2)}{(s+1)(s+3)}$ (b) $F(s) = \frac{s(s+1)}{(s+2)^2(s+3)}$	CO 2, CO3, CO5

	OR	
(c)	Derive the necessary conditions for a two-port network to be reciprocal and symmetric, expressed in terms of the impedance (Z) parameters.	CO6, CO1, CO2
Q. 4	(2×10 Marks)	
(a)	Obtain ABCD parameters for the network shown in Fig 	CO6, CO1, CO2
	OR	
(b)	Define a two-port network. Explain the terms reciprocity and symmetry. Also, write the expressions of different types of two-port network parameters.	CO6, CO1, CO2
	OR	
(c)	Analyse the single-phase parallel RL and parallel RC circuits supplied by a sinusoidal voltage. For each circuit obtain: (i) expressions for branch currents and total current, (ii) admittance and equivalent impedance, (iii) phasor diagram, (iv) power triangle, and (v) impedance triangle.	CO1, CO2, CO4
Q. 5	(2×10 Marks)	
(a)	A series RL circuit in which R= 5 ohm and L= 20 mH has an applied voltage v =100 + 50 sin wt +25 sin3wt, with w =500 rad/s. Find the current and the average power. Compute the equivalent impedance of the circuit at each frequency found in the voltage function. Then obtain the respective currents.	CO1, CO2, CO4
	OR	
(b)	Find the average power supplied to a network if the applied voltage and resulting current are $v = 50 + 50\sin 5 \times 10^3 t + 30\sin 10^4 t + 20\sin 2 \times 10^4 t$ (V) $i = 11.2\sin(5 \times 10^3 t + 63.4^\circ) + 10.6\sin(10^4 t + 45^\circ) + 8.97\sin(2 \times 10^4 t + 26.6^\circ)$ (A)	CO1, CO2, CO4
	OR	
(c)	A series RL circuit with R = 10 ohms and L = 5H contains a current $i(t) = 10\sin 1000t + 5\sin 3000t + 3\sin 5000t$ A Find the effective voltage and the average power	CO1, CO2, CO4